Patent Figures: DINOv2 with Registers Embedding Analysis

Images: 663 aircraft from 572 patents (1 639-patent dataset, labelling closed 2026-09-19, domain tags of 2026-09-22)
Label: G1 topType, the codebook's 12 architecture types (human label)
Model: `facebook/dinov2-with-registers-large` (4 register tokens), frozen; 518 px; layers 18, 22, 24 × CLS token / patch-token mean; embedding tag `dinov2-reg-large_518`
Date: 2026-09-30

Registers twin of `eVTOL-Visual-Evaluation/docs/embedding_evaluation/PATENT_EMBEDDING_ANALYSIS.pdf` (`facebook/dinov2-large`, tag `dinov2-large_518`): same images, labels, protocol and code, only the checkpoint differs; read the two side by side. Companion report: `EVTOLNEWS_DS/3_embedding_evaluation/embedding_analysis_registers/EVTOLNEWS_EMBEDDING_ANALYSIS_REGISTERS.pdf` (the same model on the evtol.news photos).

Summary

question	result	status
Are the images the right ones? (Section 1)	human labels; G1 matches the whole-patent reading for 627/662; same processed figures as the DINOv2-large run	PASSED
Are the embeddings sound? (T1, T2)	no invalid row; PC1 27 to 56 × random	PASSED
Do same-class aircraft sit closer? (T3)	yes, weakly: ratio up to 1.073, d up to 0.28	PASSED, small effect
Can the architecture be read? (T4)	L24 CLS: kNN-5 0.23 [0.19, 0.28], probe 0.36; chance 0.08, shuffled 0.10	PASSED, weak
Does it show without labels? (T5)	k-means ARI 0.026 (L24 CLS)	NO
What else is encoded? (T6)	view group of the drawing: d ratio 1.70 (L24 CLS)	CAUTION
Does the image choice matter? (T7)	rules span 0.17 to 0.23 (L24 CLS)	SMALL EFFECT
Which layer?	L24 CLS, wins 96 % of random subsets	STABLE

Across the two sources. On the same five classes, with the same protocol and matrix (L24 CLS), the patent figures reach a kNN-5 balanced accuracy of 0.44 [0.36, 0.53] and the evtol.news photos 0.56 [0.53, 0.60] (chance 0.20; probe 0.61 against 0.66). The same model reads the architecture from photographs and renders better than from patent drawings.

Against the base model. The same test (kNN-5 balanced accuracy, maker held out, exp3 main figure) with DINOv2-large with registers and with DINOv2-large; the base numbers are read from the base report's tables (`1639_LABELLED/3_embedding_evaluation/embedding_protocol`).

DINOv2-large and DINOv2-large with registers on the same aircraft, 12 classes.

matrix	kNN-5, DINOv2-large	kNN-5, DINOv2-large with registers	difference	probe, DINOv2-large	probe, DINOv2-large with registers
L18 CLS	0.097	0.118	0.021	0.162	0.210
L18 patch mean	0.091	0.106	0.015	0.241	0.238
L22 CLS	0.163	0.176	0.013	0.307	0.351
L22 patch mean	0.119	0.140	0.021	0.287	0.292
L24 CLS	0.178	0.235	0.057	0.319	0.362
L24 patch mean	0.183	0.166	-0.016	0.258	0.302

DINOv2-large and DINOv2-large with registers on the same aircraft, five parent classes.

matrix	kNN-5, DINOv2-large	kNN-5, DINOv2-large with registers	difference	probe, DINOv2-large	probe, DINOv2-large with registers
L18 CLS	0.198	0.221	0.023	0.377	0.424
L18 patch mean	0.215	0.216	0.000	0.408	0.458
L22 CLS	0.260	0.287	0.026	0.540	0.597
L22 patch mean	0.224	0.236	0.012	0.507	0.488
L24 CLS	0.333	0.445	0.111	0.568	0.606
L24 patch mean	0.339	0.339	0.000	0.531	0.501

The layer rule (Section 8) carries forward L24 CLS here and L24 CLS in the base report. A paired bootstrap of the L24 CLS difference (same aircraft, same neighbours) is in `1639_LABELLED/3_embedding_evaluation/registers_check/registers_paired.csv`.

How to read this report. Section 1 is specific to this source. Sections 2 to 8 apply one protocol, with the same code, parameters and wording, to the patent figures and to the evtol.news photos, each in its own report; a table number in one report answers the same question as in the other. The unit is the aircraft (one vector each). Every test that compares aircraft leaves out pairs and neighbours of the same maker. Six matrices are compared: DINOv2-large with registers (frozen), layers 18, 22 and 24, each as the CLS token and as the mean of the patch tokens, extracted at 518 px. The table numbers are also those of the base report of the same source.

Section 1: Which Figures Are Analysed, and Is Each Step Sound?

> **What this section establishes:** how the patent figures became one vector per aircraft, which setting each step uses and how it was chosen, and the check that shows the step does not decide the result.

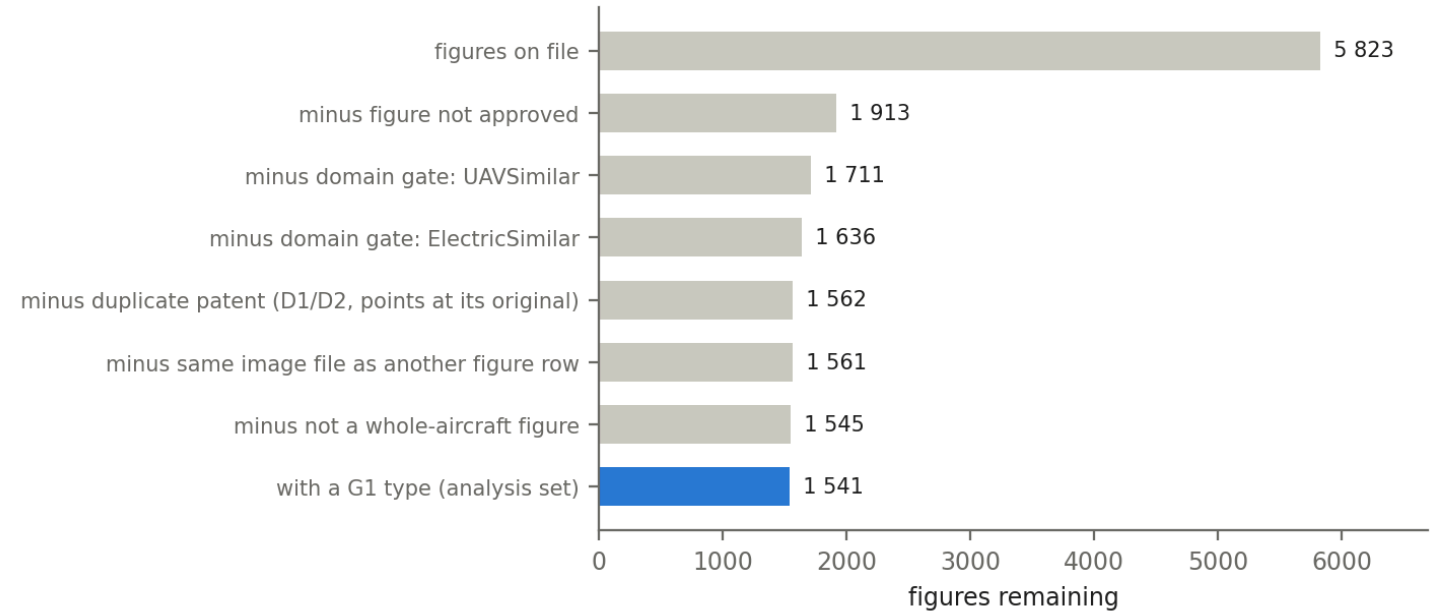
The unit is the **aircraft** (`<patent>_ua<N>`): one patent can disclose several aircraft, and the G1 type and the main-figure marker are recorded per aircraft. Every label below is a human label from the labelling wizard (Stage 04 tables of 2026-09-30); no model chooses a figure.

Registers twin. This report repeats `eVTOL-Visual-Evaluation/docs/embedding_evaluation/PATENT_EMBEDDING_ANALYSIS.pdf` with one change: the embeddings come from `facebook/dinov2-with-registers-large` (4 register tokens, embedding tag `dinov2-reg-large_518`) instead of `facebook/dinov2-large` (tag `dinov2-large_518`). The images, the labels, the selection, the processing and the protocol are the same files and the same code, so every difference between the two reports comes from the checkpoint. Section 1 is therefore the same as in the base report except for the extraction step.

1.1 The chain, step by step

Table 1. The selection chain of the patent figures and the check of each step.

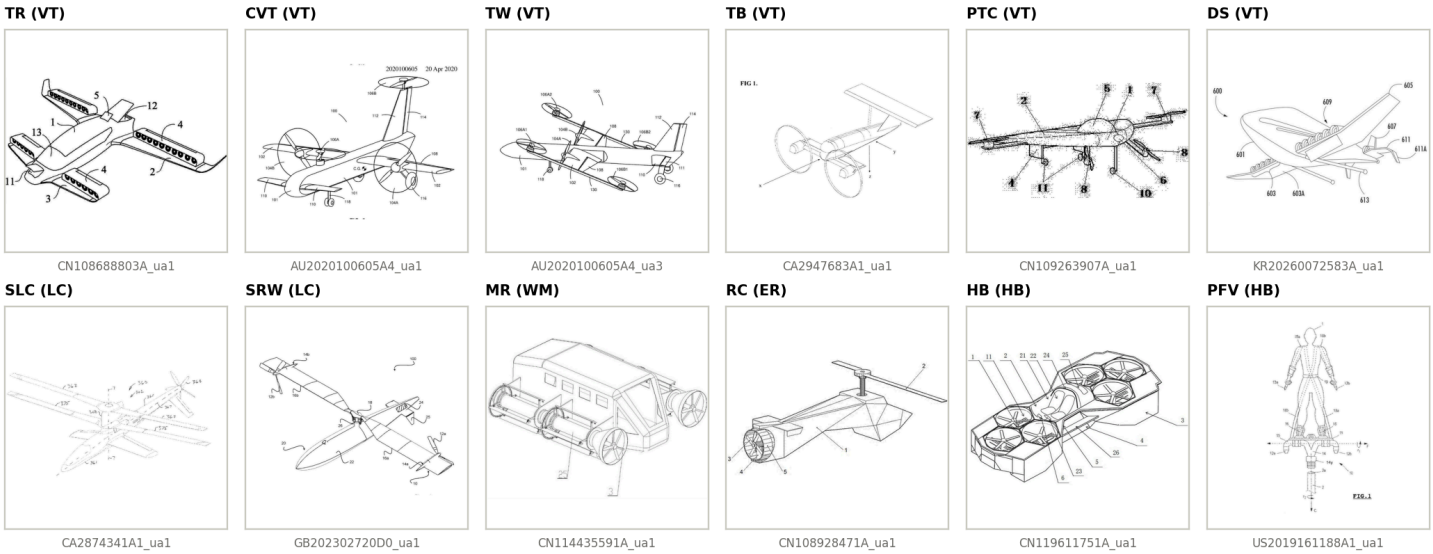
step	what it does	setting and how it was chosen	check
1 Approved figures	keep the figures the labeller approved, of approved in-domain aircraft	human approval in the wizard	every approved figure has its file on disk (0 missing)
2 Domain gate	drop aircraft tagged UAV-similar, not electric (ElectricSimilar) or STOL-only	the Preliminary Analysis gate (user ruling 2026-09-15; tags as of 2026-09-22)	same 665 aircraft as the Preliminary Analysis
3 Duplicates	drop D1/D2 duplicate patents (they repeat an original); keep D3 variants	codebook duplicate rules	each D1/D2 points at an existing original (notebook 04 check)
4 Whole aircraft only	drop part and detail figures	the wizard's parts field	16 part figures removed; 0 whole-aircraft figures without a view
5 G1 label	keep aircraft with a G1 type; 2 marked unclassifiable are left out	human G1 label	agrees with the whole-patent reading for 627 of 662 aircraft (κ 0.94)
6 One figure per aircraft	four rules pick the figure(s); the main figure is the primary rule	fixed orderings of view and flight state (notebook 23)	Section 7: the score under every rule
7 Processing	rotate by the labelled angle, pad to a white square, resize to 518 px	notebook 21, same code for both sources	0 processed images older than their source
8 Extraction	DINOv2-large with registers, frozen, layers 18/22/24, CLS token and patch-token mean	notebook 22's code, run with `facebook/dinov2-with-registers-large` on 2026-09-30 (eVTOL-Embedding-Extraction/scripts/registers_experiment.py)	1545 figures, the same figures, order and processed images as the DINOv2-large run



Source: patent figures; notebook 20 funnel (selection/funnel.csv) and notebook 23 figure table; each figure counted under the first rule that removed it

Figure 1. From figures on file to the analysis set.
Source: patent figures; notebook 20 funnel (selection/funnel.csv) and notebook 23 figure table; each figure counted under the first rule that removed it.

The analysis set holds **663 aircraft from 572 patents** and 1541 whole-aircraft figures. 74 figures leave with duplicate patents, 277 with the domain gate.



Source: patent figures, rule exp3 main figure; the alphabetically first aircraft of each type whose main figure is a Perspective view; label = G1 type (its five-class parent)

Figure 2. One main figure per G1 type, as the model receives it (518 px, padded square).
Source: patent figures, rule exp3 main figure; the alphabetically first aircraft of each type whose main figure is a Perspective view; label = G1 type (its five-class parent).

1.2 The labels the rules sort on

View. The eight wizard views are grouped into four: Perspective (front- and rear-isometric), Plan (top, bottom), Side, Front/Rear. **Flight state** matters only for a convertible type (TW, TR, CVT, DS, SRW), whose shape changes between hover and cruise. For the fixed types every figure is Invariant. On a convertible aircraft Hover and Cruise are kept; a

figure labelled Invariant (it fits both states) or Both (the moving part drawn in both positions, added to the wizard on 2026-09-19) becomes **Both**; Transition and Other become Other; an unlabelled figure is Missing.

Table 2. Flight state of each whole-aircraft figure, by G1 type.

G1	Cruise	Both	Hover	Other	Invariant	figures
TR	189	11	197	32	0	429
CVT	128	10	133	14	0	285
TW	77	4	79	7	0	167
TB	0	0	0	0	57	57
PTC	0	0	0	0	41	41
DS	9	0	3	2	0	14
SLC	0	0	0	0	346	346
SRW	11	6	4	1	0	22
MR	0	0	0	0	106	106
RC	0	0	0	0	17	17
HB	0	0	0	0	34	34
PFV	0	0	0	0	23	23

Views: Perspective 822, Plan 323, Side 252, Front/Rear 144. The canonical state for the state-first rule is Cruise, re-checked by notebook 23 on today's labels.

1.3 One vector per aircraft: the four rules

Table 3. The four image rules (notebook 23).

rule	which figure
exp1 view-first	best view (Perspective > Plan > Side > Front/Rear), then best flight state
exp2 state-first	best flight state (Cruise > Both > Hover > Other > Missing), then best view
exp3 main figure	the figure the labeller marked as main
exp4 view average	mean of the best figure in each of Perspective, Plan and Side

Table 4. How often two single-figure rules pick the same figure.

pair	same figure	aircraft that differ
exp1 vs exp2	96.1 %	26
exp1 vs exp3	83.9 %	107
exp2 vs exp3	83.4 %	110

exp4 averages 1 slot for 416 aircraft (then it equals one figure), 2 for 197 and 3 for 50. 1 aircraft whose main figure is a part figure falls back to exp1's pick (US6367736B1_ua1).

1.4 Does the selection decide the result?

The primary score of Section 4 is recomputed after each selection step is made stricter: only the aircraft whose drawing label matches the whole-patent reading (label noise removed), only the aircraft whose main figure is a Perspective view (view held constant), and without the fallbacks.

Table 5. Sensitivity of the kNN-5 score to the selection (L24 CLS, maker held out).

variant	aircraft	kNN-5	95 % interval
as built	663	0.235	0.19 to 0.28
only aircraft whose drawing label matches the whole-patent reading	627	0.241	0.20 to 0.28
only aircraft whose main figure is a Perspective view	477	0.244	0.18 to 0.31
without the main-figure fallbacks	662	0.235	0.19 to 0.28

Every variant stays inside the interval of the full set: no selection step decides the result.

SELECTION: HUMAN LABELS, CHECKED

Section 2: Are the Embeddings Sound? (T1, T2)

> **What this section checks:** that every matrix is valid and carries more structure than random noise. Nothing here uses the architecture label.

Table 6. Integrity and label-free structure, one vector per aircraft (exp3 main figure, 663 aircraft).

matrix	cosine mean	cosine SD	PC1 ÷ random	effective dims	dims for 90 %	Hopkins
L18 CLS	0.928	0.035	46.8	12.9	73	0.714
L18 patch mean	0.927	0.032	37.1	15.2	84	0.736
L22 CLS	0.755	0.056	27.4	37.0	193	0.702
L22 patch mean	0.931	0.030	36.1	16.1	90	0.731
L24 CLS	**0.487**	**0.129**	**32.2**	**26.5**	**181**	**0.742**
L24 patch mean	0.864	0.062	56.3	10.1	66	0.710

Integrity: no row with a NaN, an infinite value or all zeros in the six matrices, and no exact duplicate row. On its first principal component every matrix carries 27 to 56 times the variance of a random matrix of the same shape (pass bar 3). Hopkins lies between 0.70 and 0.74: the vectors clump, but as a continuum and not as separate islands (0.5 would be uniform).

T1 AND T2: PASSED

Section 3: Do Same-Class Aircraft Sit Closer Together? (T3)

> **What this section checks:** whether two aircraft of the same class are, on average, closer in the embedding than two aircraft of different classes.

Separation ratio = mean cosine distance between aircraft of different classes ÷ mean distance between aircraft of the same class; 1.0 means the class is invisible. **Cohen's d** is the same gap in pooled standard deviations. Pairs of the same maker are left out, so a company's house style cannot count as architecture. p from 999 label permutations.

Table 7. Class separation, exp3 main figure, label: G1 topType, the codebook's 12 architecture types (human label).

matrix	ratio	Cohen's d	p	pairs same class	pairs different class
L18 CLS	1.029	0.060	0.027	37 971	178 869
L18 patch mean	1.031	0.068	0.011	37 971	178 869
L22 CLS	1.045	0.194	≤ 0.001	37 971	178 869
L22 patch mean	1.046	0.102	≤ 0.001	37 971	178 869
L24 CLS	**1.073**	**0.277**	**≤ 0.001**	**37 971**	**178 869**
L24 patch mean	1.038	0.081	0.003	37 971	178 869

The ratio lies between 1.029 and 1.073, and 6 of 6 matrices separate the classes at p ≤ 0.05. The effect is small in every matrix (largest d 0.28, L24 CLS): the classes overlap heavily on average. Averages over all pairs hide local structure, which the next test measures.

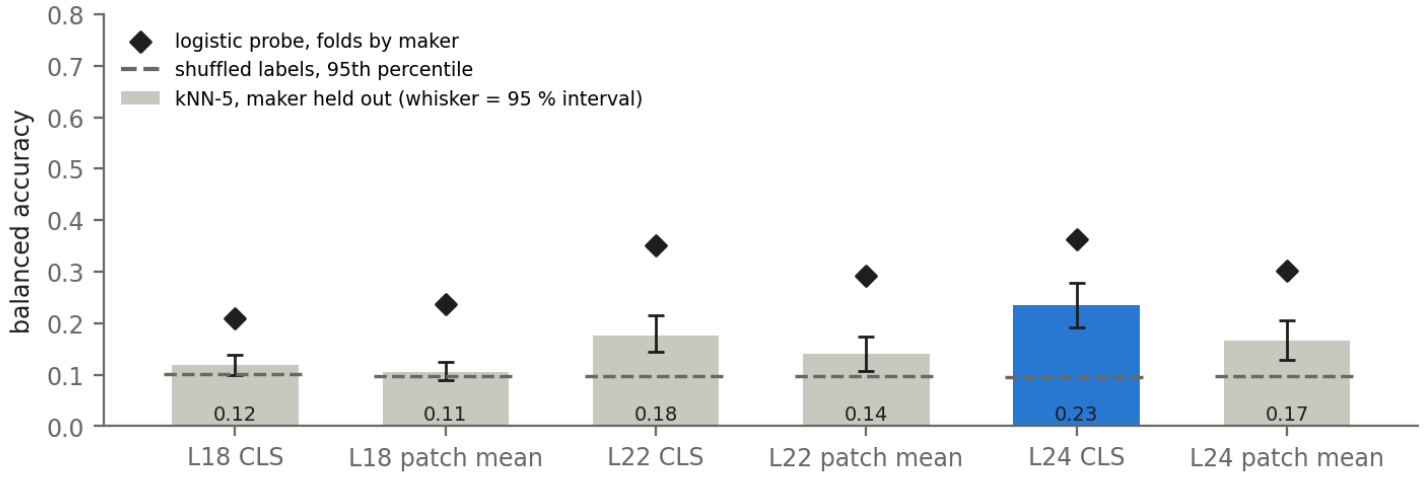
Section 4: Can the Architecture Be Read from the Embedding? (T4)

> **What this section checks:** how well the class of an aircraft can be predicted from the classes of its nearest neighbours, and by a linear probe, with the whole maker held out.

kNN-5 (primary score). The class of each aircraft is predicted by the majority class of its five nearest aircraft (cosine), never counting an aircraft of the same maker (the applicant's company group, or the first applicant's name for individual inventors). The score is the **balanced accuracy**, the mean recall over the classes, so that a large class cannot carry it; chance is 1/12 = 0.08. The interval is a 95 % bootstrap over the aircraft. **Shuffled labels** give the chance level actually reached by the same neighbours (95th percentile of 200 shuffles). **Probe**: logistic regression, 5 folds grouped by maker.

Table 8. Prediction of the architecture, exp3 main figure, maker held out (663 aircraft, 12 classes).

matrix	kNN-5	95 % interval	macro-F1	shuffled, p95	probe
L18 CLS	0.118	0.10 to 0.14	0.117	0.101	0.210
L18 patch mean	0.106	0.09 to 0.12	0.106	0.098	0.238
L22 CLS	0.176	0.14 to 0.22	0.194	0.098	0.351
L22 patch mean	0.140	0.11 to 0.17	0.148	0.098	0.292
L24 CLS	**0.235**	**0.19 to 0.28**	**0.234**	**0.096**	**0.362**
L24 patch mean	0.166	0.13 to 0.20	0.181	0.096	0.302



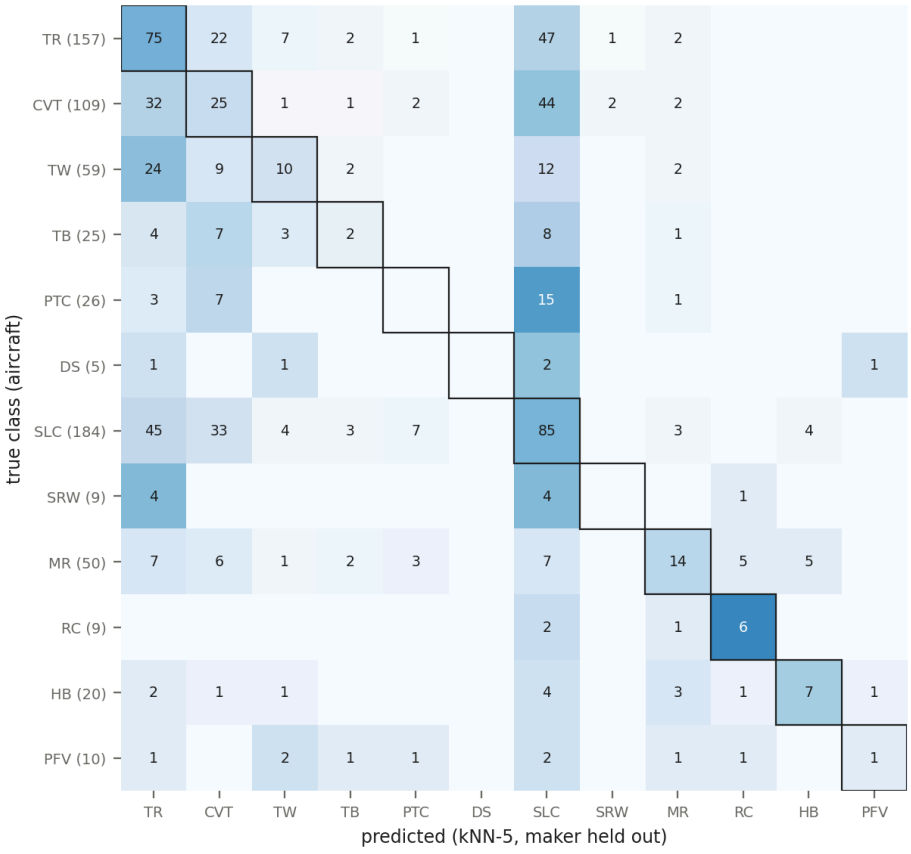
Source: patent figures; DINOv2-large with registers, frozen, notebook 22 extraction code at 518 px; layers 18/22/24 x CLS token / patch-token mean; image rule: exp3 main figure; 663 aircraft; label: G1 topType, the codebook's 12 architecture types (human label)
How to read: balanced accuracy = mean recall over the classes, so a large class cannot carry the score; chance = 1/12

Figure 3. Architecture read from the embedding, per layer and pooling (highlighted: the matrix the layer rule selects).
Source: patent figures; DINOv2-large with registers, frozen, notebook 22 extraction code at 518 px; layers 18/22/24 x CLS token / patch-token mean; image rule: exp3 main figure; 663 aircraft; label: G1 topType, the codebook's 12 architecture types (human label).
How to read: balanced accuracy = mean recall over the classes, so a large class cannot carry the score; chance = 1/12.

L24 CLS scores 0.23 (interval 0.19 to 0.28) against a shuffled level of 0.10; the probe reaches 0.36. In 4 of 6 matrices the whole interval lies above the shuffled level. The probe, which weighs every dimension, reads the classes better than the neighbours do (0.36 against 0.23): the architecture is present in the vector, but it is not what decides which aircraft are nearest.

Table 9. Recall per class, L24 CLS, kNN-5 with the maker held out.

class	aircraft	recall
TR	157	0.48
CVT	109	0.23
TW	59	0.17
TB	25	0.08
PTC	26	0.00
DS	5	0.00
SLC	184	0.46
SRW	9	0.00
MR	50	0.28
RC	9	0.67
HB	20	0.35
PFV	10	0.10



Source: patent figures; DINOv2-large with registers, frozen, notebook 22 extraction code at 518 px; layer 24, CLS token; image rule: exp3 main figure; 663 aircraft
How to read: rows = true class; columns = the class kNN predicts; numbers = aircraft; the diagonal is correct; darker = larger share of the row

Figure 4. Which classes are confused with which.

Source: patent figures; DINOv2-large with registers, frozen, notebook 22 extraction code at 518 px; layer 24, CLS token; image rule: exp3 main figure; 663 aircraft.

How to read: rows = true class; columns = the class kNN predicts; numbers = aircraft; the diagonal is correct; darker = larger share of the row.

The same test on the five parent classes. The evtol.news report can only use the directory's five classes. For a like-for-like comparison, each G1 type is mapped to its parent (TR, CVT, TW, TB, PTC, DS → Vectored Thrust; SLC, SRW → Lift + Cruise; MR → Wingless; RC → Electric Rotorcraft; HB, PFV → Hover Bikes / PFD) and the test is repeated.

Table 10. Prediction of the five parent classes, same protocol.

matrix	kNN-5	95 % interval	shuffled, p95	probe
L18 CLS	0.221	0.20 to 0.25	0.215	0.424
L18 patch mean	0.216	0.20 to 0.23	0.219	0.458
L22 CLS	0.287	0.25 to 0.33	0.216	0.597
L22 patch mean	0.236	0.20 to 0.29	0.216	0.488
L24 CLS	**0.445**	**0.36 to 0.53**	**0.217**	**0.606**
L24 patch mean	0.339	0.27 to 0.41	0.219	0.501

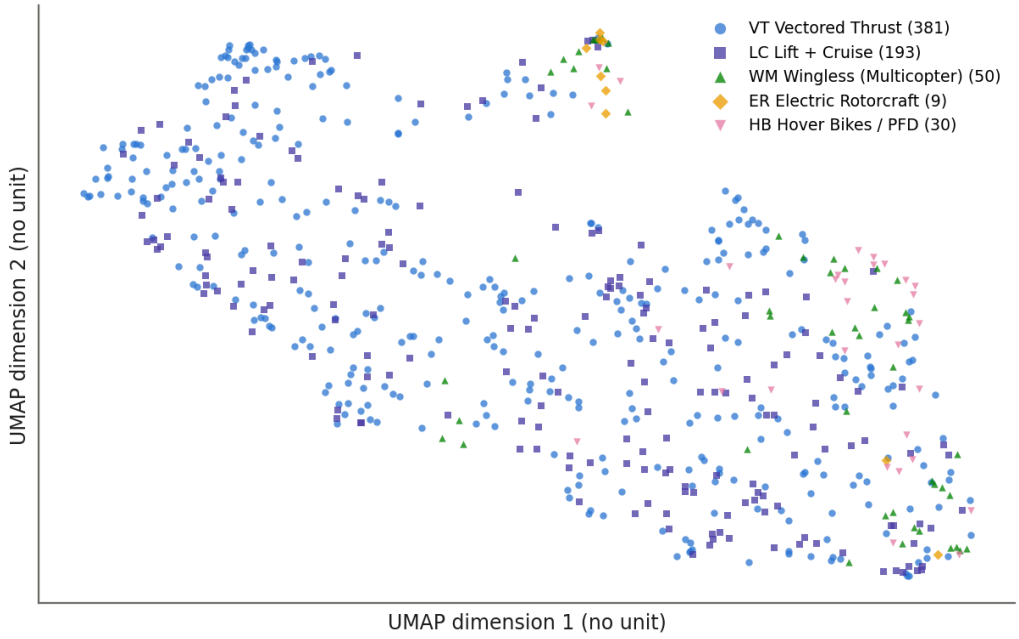
On five classes L24 CLS reaches 0.44 (chance 0.20).

Section 5: Does the Structure Follow the Classes Without Labels? (T5)

> **What this section checks:** whether a partition found without any label (k-means with as many clusters as classes) matches the classes.

Table 11. k-means (12 clusters) against the classes. ARI 0 = chance, 1 = identical.

matrix	ARI	NMI
L18 CLS	0.031	0.103
L18 patch mean	0.011	0.073
L22 CLS	0.033	0.143
L22 patch mean	0.014	0.080
L24 CLS	**0.026**	**0.139**
L24 patch mean	0.012	0.083



Source: patent figures; DINOv2-large with registers, frozen, notebook 22 extraction code at 518 px; layer 24, CLS token; image rule: exp3 main figure; 663 aircraft; UMAP n_neighbors 15, min_dist 0.1, cosine, seed 42
How to read: each point is one aircraft. UMAP reduces each 1 024-number embedding to 2 numbers so that aircraft with similar embeddings sit close together. The two axes are these 2 numbers: no unit, no physical meaning, arbitrary orientation. Only closeness is meaningful; positions on the axes and distances between far-apart groups are not

Figure 5. The embedding space of the selected matrix, coloured by the five-class parent of the G1 type.
Source: patent figures; DINOv2-large with registers, frozen, notebook 22 extraction code at 518 px; layer 24, CLS token; image rule: exp3 main figure; 663 aircraft; UMAP n_neighbors 15, min_dist 0.1, cosine, seed 42.
How to read: each point is one aircraft. UMAP reduces each 1 024-number embedding to 2 numbers so that aircraft with similar embeddings sit close together. The two axes are these 2 numbers: no unit, no physical meaning, arbitrary orientation. Only closeness is meaningful; positions on the axes and distances between far-apart groups are not.

ARI lies between 0.011 and 0.033. The label-free clusters do not follow the classes: whatever structure the space has, it is not organised by architecture first.

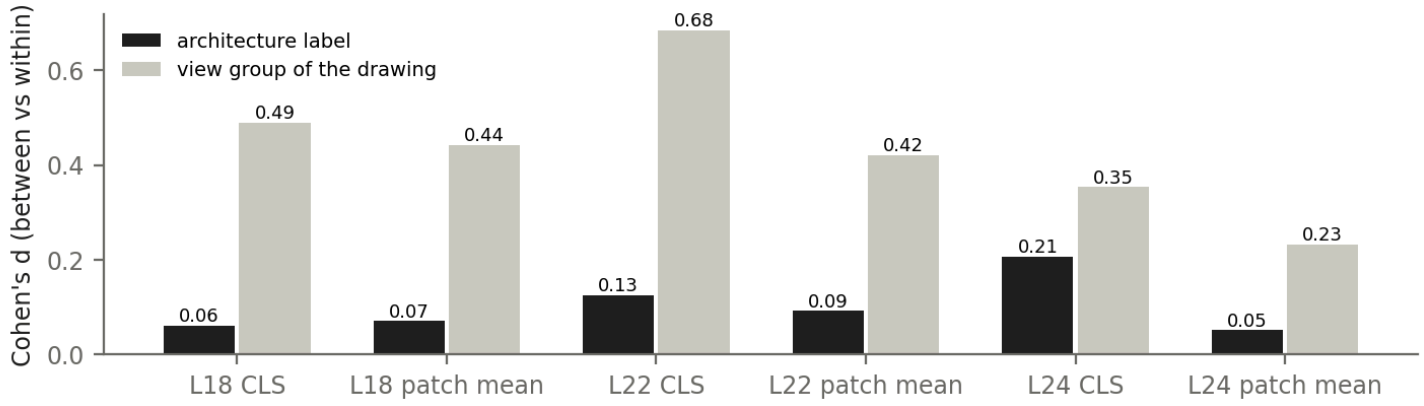
Section 6: What Else Does the Embedding Encode? (T6)

> **What this section checks:** whether a property of the image that is not the architecture (view group of the drawing) separates the vectors more strongly than the architecture does.

The T3 separation is computed twice on the same vectors and the same pairs (all 1541 whole-aircraft figures of the 663 aircraft; same-maker pairs left out): once with the architecture as the label, once with the view group of the drawing (Perspective, Plan, Side, Front/Rear). A ratio d(confound) ÷ d(architecture) above 1 means that property moves the vectors more than the architecture.

Table 12. Architecture against view group of the drawing.

matrix	architecture d	view d	view d ÷ architecture d
L18 CLS	0.061	0.489	8.04
L18 patch mean	0.070	0.443	6.31
L22 CLS	0.125	0.685	5.47
L22 patch mean	0.092	0.421	4.57
L24 CLS	**0.207**	**0.353**	**1.70**
L24 patch mean	0.051	0.232	4.52



Source: patent figures; DINOv2-large with registers, frozen, notebook 22 extraction code at 518 px; layers 18/22/24 x CLS token / patch-token mean; 663 aircraft; all 1541 whole-aircraft figures of the 663 aircraft; pairs of the same maker left out
How to read: d = how much further apart two items of different labels sit than two items of the same label, in standard deviations; a taller grey bar means this label moves the vectors more than the architecture

Figure 6. What else the embedding encodes: the view group of the drawing against the architecture.

Source: patent figures; DINOv2-large with registers, frozen, notebook 22 extraction code at 518 px; layers 18/22/24 x CLS token / patch-token mean; 663 aircraft; all 1541 whole-aircraft figures of the 663 aircraft; pairs of the same maker left out.

How to read: d = how much further apart two items of different labels sit than two items of the same label, in standard deviations; a taller grey bar means this label moves the vectors more than the architecture.

In 6 of 6 matrices the view group of the drawing separates the vectors more than the architecture (ratio 1.70 to 8.04; L24 CLS: 1.70).

Section 7: Does the Choice of Image Matter? (T7)

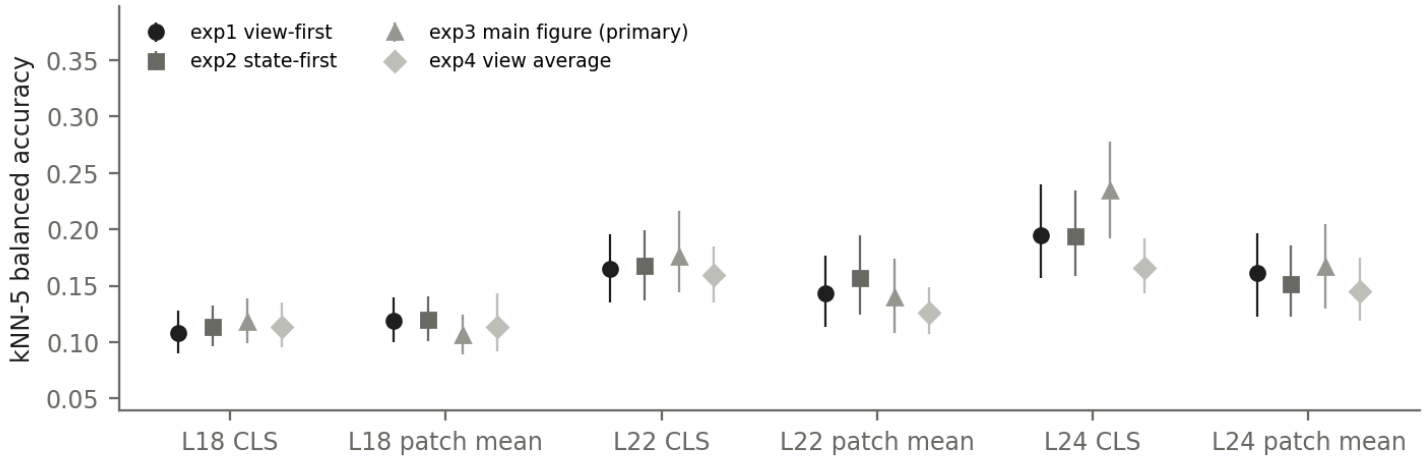
> **What this section checks:** whether the result depends on which figure of each aircraft is used. The T4 score is recomputed under every image rule.

Table 13. The image rules.

rule	which image
exp1 view-first	best view (Perspective > Plan > Side > Front/Rear), then best flight state
exp2 state-first	best flight state (Cruise > Both > Hover > Other > Missing), then best view
exp3 main figure	the figure the labeller marked as main
exp4 view average	mean of the best figure in each of Perspective, Plan and Side

Table 14. kNN-5 balanced accuracy (maker held out) per image rule.

matrix	exp1 view-first	exp2 state-first	exp3 main figure	exp4 view average
L18 CLS	0.108	0.113	0.118	0.113
L18 patch mean	0.119	0.120	0.106	0.113
L22 CLS	0.165	0.168	0.176	0.160
L22 patch mean	0.143	0.157	0.140	0.126
L24 CLS	**0.195**	**0.194**	**0.235**	**0.166**
L24 patch mean	0.161	0.152	0.166	0.145



Source: patent figures; DINOv2-large with registers, frozen, notebook 22 extraction code at 518 px; layers 18/22/24 x CLS token / patch-token mean; 663 aircraft
How to read: one marker per image rule; vertical line = 95 % bootstrap interval

Figure 7. Does the choice of image change the result? The same test under every image rule.
Source: patent figures; DINOv2-large with registers, frozen, notebook 22 extraction code at 518 px; layers 18/22/24 x CLS token / patch-token mean; 663 aircraft.
How to read: one marker per image rule; vertical line = 95 % bootstrap interval.

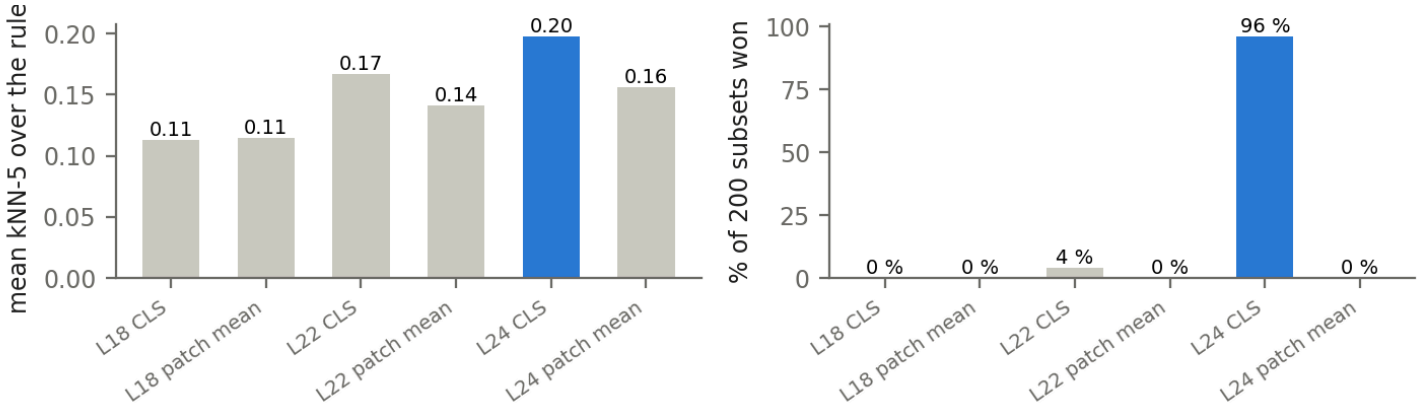
On L24 CLS the rules range from 0.17 to 0.23, a spread of 0.07, against a 95 % interval about 0.09 wide for one rule. The highest score comes from exp3 main figure. The spread is smaller than the interval, so no rule changes the conclusion, but the order of the rules is consistent enough to note.

Section 8: Which Layer? (Layer Rule)

> **Rule, fixed before the numbers were read:** the matrix with the highest kNN-5 score (maker held out) averaged over the image rules is carried forward. Robustness: the same choice is repeated on 200 random subsets of 80 % of the aircraft.

Table 15. Layer rule.

matrix	mean kNN-5 over the rules	subsets won
L18 CLS	0.113	0 %
L18 patch mean	0.115	0 %
L22 CLS	0.167	4 %
L22 patch mean	0.141	0 %
L24 CLS	**0.197**	**96 %**
L24 patch mean	0.156	0 %



Source: patent figures; DINOv2-large with registers, frozen, notebook 22 extraction code at 518 px; layers 18/22/24 x CLS token / patch-token mean; 663 aircraft

Figure 8. Layer rule: the matrix with the best mean score, and how often it stays best when the aircraft change.
Source: patent figures; DINOv2-large with registers, frozen, notebook 22 extraction code at 518 px; layers 18/22/24 x CLS token / patch-token mean; 663 aircraft.

L24 CLS is carried forward. It wins 96 % of the subsets. The choice is stable.

`LAYER: L24 CLS`

Open Points

1. **Small classes.** DS 5, SRW 9, RC 9, PFV 10 have fewer than 20 aircraft; their recall rests on a handful of aircraft (Table 9).
2. **Wizard.** Mark a whole-aircraft main figure for US6367736B1_ua1; check whether KR20240170008A ua1 and ua2 are one aircraft drawn in two states.
3. **Notebook 23** still writes its experiment folders from L22 CLS (`PRIMARY_LAYER`); this report reads the full matrices and does not depend on it.

Generated by `embedding_evaluation/scripts/build_registers_reports.py`, which runs `build_embedding_reports.py` on the tag `dinov2-reg-large_518`, with `src/embedding_protocol.py` (the shared protocol) and `src/embedding_reports.py`. Sources: `1639_LABELLED/2_embedding_extraction/`: `selection/` (notebook 20), `processed/518/` (21), `embeddings/dinov2-reg-large_518/` (22, run by registers_experiment.py), `view_state_experiments/` (23); labels `0_labelling/outputs/` (notebook 04). Tables as CSV in `1639_LABELLED/3_embedding_evaluation/embedding_protocol_registers/`.